

Clustering

Faculty of Information Technology, Monash University, Australia

FIT5149 week 6

- 1 K-Means Clustering
- 2 Hierarchical Clustering
- 3 Clustering Learned Representations
- 4 Summary

- Weekly learning outcomes
 - ▶ Understand the clustering methods introduced in this unit
 - ▶ Be able to apply appropriate clustering methods in practice
 - ▶ Understand how these methods are applied to learned representations

K-Means Clustering

- The basic idea: partition a data set into K **distinct, non-overlapping** clusters and minimise the within-cluster variation.
- The within-cluster variation for cluster C_k : a measure $W(C_k)$ of the amount by which the observations within a cluster differ from each other.
- With Euclidean distance,

$$W(C_k) = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2$$

- The problem to be solved (**global optimum**):

$$\text{minimise}_{C_1, \dots, C_K} \left\{ \sum_{k=1}^K W(C_k) \right\}$$

- The number of possible partitions grows exponentially in n , so we do not solve this exactly. We use Lloyd's greedy algorithm instead [2].

K-Means Clustering: Algorithm

Algorithm: K -Means Clustering

1. Randomly assign a number, from 1 to K , to each of the observations. These serve as initial cluster assignments for the observations.
2. Iterate until the cluster assignments stop changing:
 - (a) For each of the K clusters, compute the cluster *centroid*. The k th cluster centroid is the vector of the p feature means for the observations in the k th cluster.
 - (b) Assign each observation to the cluster whose centroid is closest (where *closest* is defined using Euclidean distance).

Interactive visualization of k -means (run live)

Properties of the algorithm

- The algorithm is guaranteed to decrease the value of the objective function at each step. Why?
 - ▶ Step 2 (a) Compute centroids. The pairwise form is equivalent to a distance-to-centroid form:

$$\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=1}^p (x_{ij} - \bar{x}_{kj})^2 \quad (1)$$

where $\bar{x}_{kj} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}$ is the mean for feature j in cluster C_k , and the mean minimises the right-hand side.

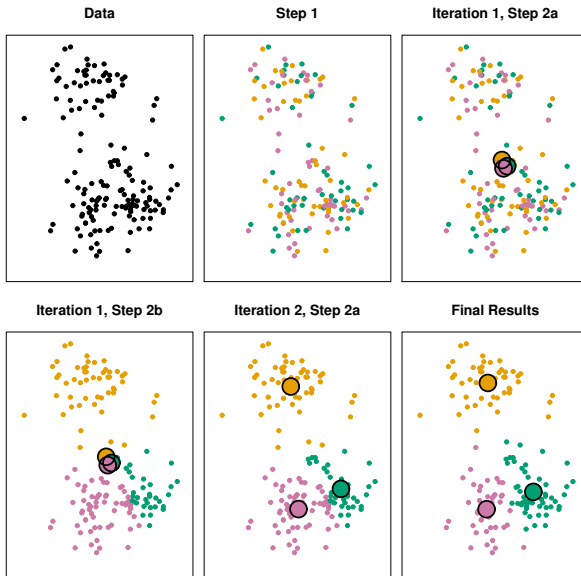
- ▶ Step 2 (b) Assign each observation to its closest centroid

$$\sum_{j=1}^p (x_{ij} - c_{kj})^2 \leq \sum_{j=1}^p (x_{ij} - c_{k'j})^2$$

where $\mathbf{c}_k$ denotes the centroid with the smallest distance.

- ▶ Both steps decrease the objective, so the algorithm converges. **However it reaches only a local minimum.**

K-Means Clustering: Example of running the algorithm



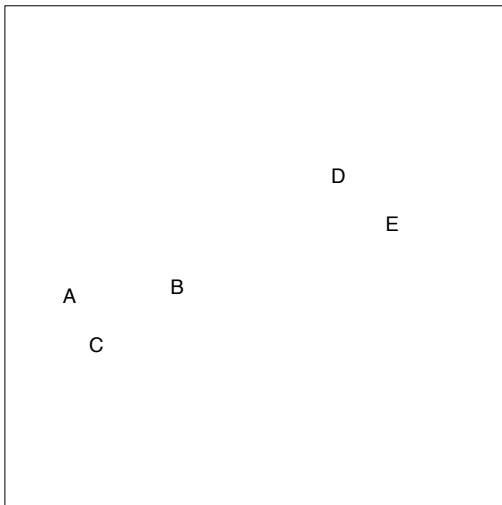
K-Means Clustering: Impact of Initialisation



- Different initialisations reach different local minima. In practice: use many restarts (`n_init`) and keep the lowest objective, or use *k*-means++ seeding [3].
- Same demo as before: restart it with different seeds to reproduce these outcomes.

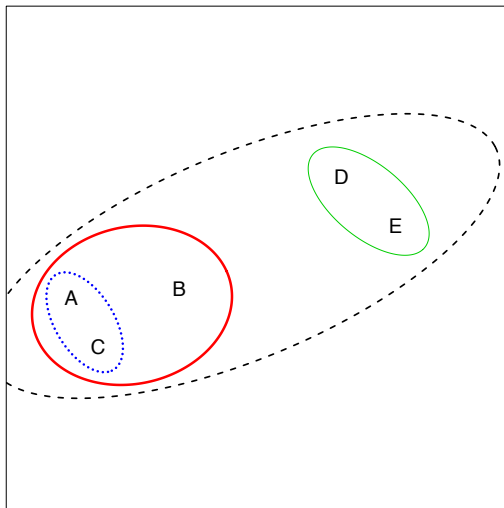
Hierarchical Clustering: Basic Idea

- Builds a hierarchy in a “bottom-up” fashion...



Hierarchical Clustering: Basic Idea

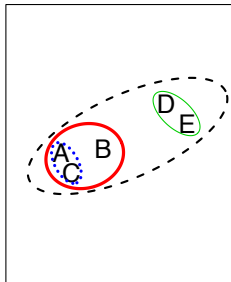
- Builds a hierarchy in a “bottom-up” fashion...



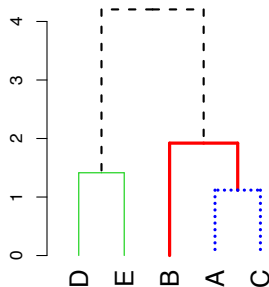
Hierarchical Clustering: Algorithm

- Steps:

- ▶ Start with each point in its own cluster
- ▶ Identify the closest two clusters and merge them
- ▶ Repeat until all points are in a single cluster

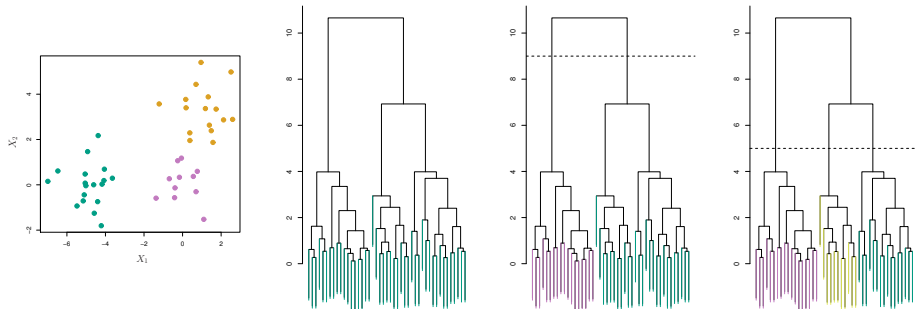


Dendrogram



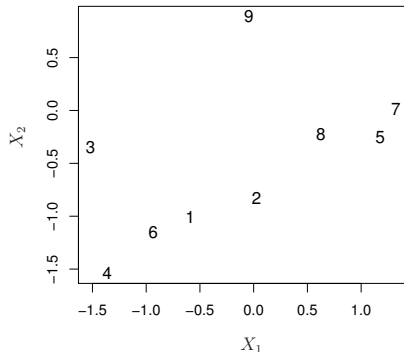
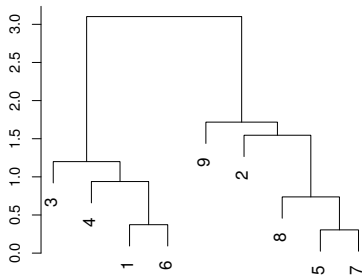
Cluster height: distance between data points at the time being clustered.

Hierarchical Clustering: Interpretation of Dendrogram



- Left: 45 observations in two dimensions with three true classes (colours), which we treat as unknown. Right: the resulting dendrogram, cut at two different heights.
- The earlier (lower in the tree) fusions occur, the more similar the groups of observations are to each other.
- Cutting the dendrogram at a given height yields a partition: the cut height, rather than K , is the tuning parameter.

Hierarchical Clustering: Interpretation of Dendrogram



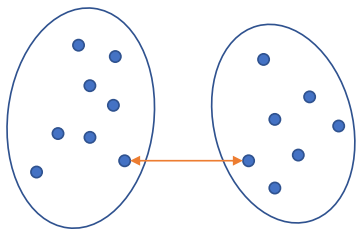
- The height of the fusion indicates how different the two observations are.
 - ▶ Incorrect: observations 9 and 2 are quite similar
 - ▶ We cannot draw conclusion about the similarity of two observations based on their proximity along the horizontal axis.

Hierarchical Clustering: Algorithm

Algorithm 10.2 *Hierarchical Clustering*

1. Begin with n observations and a measure (such as Euclidean distance) of all the $\binom{n}{2} = n(n-1)/2$ pairwise dissimilarities. Treat each observation as its own cluster.
 2. For $i = n, n-1, \dots, 2$:
 - (a) Examine all pairwise inter-cluster dissimilarities among the i clusters and identify the pair of clusters that are least dissimilar (that is, most similar). Fuse these two clusters. The dissimilarity between these two clusters indicates the height in the dendrogram at which the fusion should be placed.
 - (b) Compute the new pairwise inter-cluster dissimilarities among the $i-1$ remaining clusters.
-

Linkage function: Single linkage¹



$$L(r, s) = \min(D(x_{r,i}, x_{s,i}))$$

pros: It can separate non-elliptical shapes as long as the gap between two clusters is not small.



cons: It cannot separate clusters properly if there is noise between clusters.

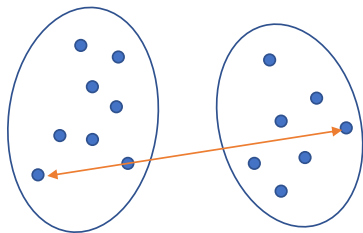


- The failure mode is *chaining*: a thin bridge of noise is enough to merge two clusters, because only the closest pair of points matters.

¹Figures: [https://towardsdatascience.com/](https://towardsdatascience.com/understanding-the-concept-of-hierarchical-clustering-technique-c6e8243758ec)

[understanding-the-concept-of-hierarchical-clustering-technique-c6e8243758ec](https://towardsdatascience.com/understanding-the-concept-of-hierarchical-clustering-technique-c6e8243758ec)

Linkage function: Complete linkage²



cons: tends to break large clusters.



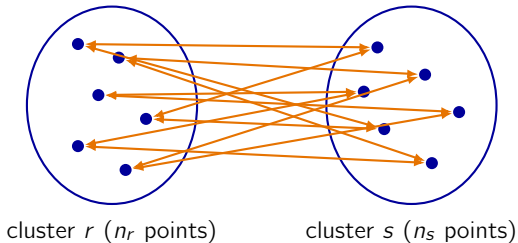
$$L(r, s) = \max(D(x_{r,i}, x_{s,i}))$$

- **Pros:** less susceptible to noise and outliers than single linkage, because a single close pair cannot force a merge.
- **Cons:** the diameter criterion penalises large clusters, so it tends to split them (right).

²Figures:

https://cse.buffalo.edu/~jing/cse601/fa12/materials/clustering_hierarchical.pdf

Linkage function: Average linkage



$$L(r, s) = \frac{1}{n_r n_s} \sum_{i=1}^{n_r} \sum_{j=1}^{n_s} D(x_{r,i}, x_{s,j})$$

- **Pros:** less susceptible to noise and outliers, since every pair contributes to the average.
- **Cons:** biased towards globular clusters.

Linkage function: Ward's Method

Ward's method [4] defines the distance between two clusters A and B as the *increase* in the within-cluster sum of squares caused by merging them:

$$\begin{aligned}\Delta(A, B) &= \sum_{i \in A \cup B} \|\mathbf{x}_i - \mathbf{m}_{A \cup B}\|^2 - \sum_{i \in A} \|\mathbf{x}_i - \mathbf{m}_A\|^2 - \sum_{i \in B} \|\mathbf{x}_i - \mathbf{m}_B\|^2 \\ &= \frac{n_A n_B}{n_A + n_B} \|\mathbf{m}_A - \mathbf{m}_B\|^2\end{aligned}$$

where $\mathbf{m}_j$ is the centre of cluster j and n_j is the number of points in it. Δ is called the *merging cost*.

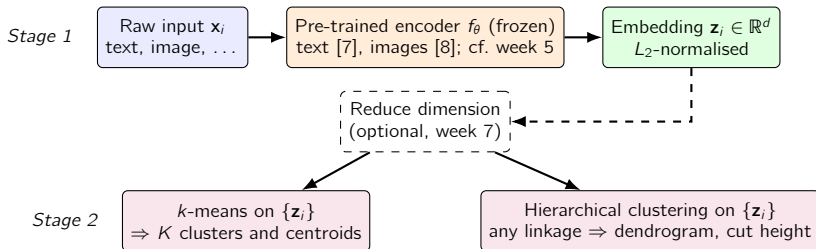
- **Ward is not centroid linkage.** The factor $\frac{n_A n_B}{n_A + n_B}$ means cluster *size* matters as well as separation.
- Given two pairs of clusters whose centres are equally far apart, Ward prefers to merge the *smaller* pair. This is why it tends to produce balanced dendrograms and roughly equal-sized, globular clusters.
- It targets the same sum-of-squares criterion as k -means, but greedily and constrained by the merges already made.

Clustering: Practical Issues

- Should the observations or features first be standardised in some way? For instance, maybe the variables should be centred to have mean zero and scaled to have standard deviation one.
- In the case of hierarchical clustering
 - ▶ What dissimilarity measure should be used?
 - ▶ What type of linkage should be used?
- How many clusters to choose?
 - ▶ Nonparametric Bayesian Methods
- And the question we turn to next: in which *space* should the distances be measured?

Embed First, Then Cluster

- Both algorithms today assume that Euclidean distance in the *input* space is meaningful. On raw pixels or raw TF-IDF counts it usually is not: distances are dominated by brightness, document length or shared stop words.
- The standard two-stage recipe: map each object to an embedding $\mathbf{z}_i = f_\theta(\mathbf{x}_i)$ with a pre-trained encoder, then cluster the embeddings with either algorithm, unchanged.



- Only the coordinates change: the k -means objective, restarts, linkage functions and dendrograms all carry over. This baseline should always be run before anything more elaborate.

Deep Embedded Clustering (DEC)

- Limitation of the two-stage recipe: the encoder was trained for some other objective, so nothing encourages the embedding to be *clusterable*.
- DEC [5] keeps the k -means picture but updates the encoder as well as the centroids.
- **Initialise.** Train an autoencoder, then run k -means on its embeddings to obtain centroids $\mu_1, \dots, \mu_K$.
- **Soft assignment.** Replace the hard assignment by a similarity to each centroid:

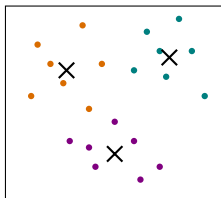
$$q_{ik} = \frac{(1 + \|\mathbf{z}_i - \mu_k\|^2)^{-1}}{\sum_{k'} (1 + \|\mathbf{z}_i - \mu_{k'}\|^2)^{-1}}$$

- **Sharpened target.** Build a target that emphasises the confident points, normalised by the soft cluster size $f_k = \sum_i q_{ik}$:

$$p_{ik} = \frac{q_{ik}^2 / f_k}{\sum_{k'} q_{ik'}^2 / f_{k'}}$$

- **Train.** Minimise $\mathcal{L} = \text{KL}(P \parallel Q) = \sum_i \sum_k p_{ik} \log \frac{p_{ik}}{q_{ik}}$ with respect to both θ and $\{\mu_k\}$.

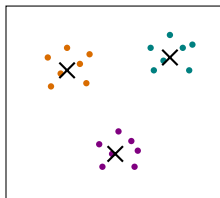
DEC: What It Buys You, and What to Watch For

Fixed embedding + k -means

q_{ik} : soft assignment
near a centroid = confident

minimise $KL(P \parallel Q)$
 $\longrightarrow$
 update θ and $\{\mu_k\}$

After DEC training



p_{ik} : sharpened target
confident points
pull the encoder

- **Pros:** the representation is adapted to the clustering task; clusters become far more compact.
- **Cons:** self-reinforcing, so a poor initialisation is amplified; K is still fixed in advance; tight is not the same as correct; gains over embedding + k -means are often modest at equal tuning budget.
- **Takeaway:** deep clustering changes the space the algorithms operate in; it does not replace them.

Demo

One dataset in the seminar notebook: **CIFAR-10**, 2,000 colour photographs of 32×32 pixels in 10 classes. Everything is scored with the adjusted Rand index [9], which unlike the Rand index is near 0 for a random partition.

- **Demo 1: raw pixels (3,072-d).**

- ▶ k -means, then the mean image of each cluster: the clusters are organised by brightness and background colour, not by object.
- ▶ Twenty seeds at `n_init=1`, random initialisation versus k -means++ [3]: how much does the objective vary, and does lower inertia always mean higher ARI?
- ▶ Read the dendrogram; cut it by K and by height.
- ▶ Ward, complete, average and single linkage. Watch the cluster sizes, not only the ARI: each linkage fails in its own characteristic way.

- **Demo 2: the same images through a frozen ResNet-18 [10] (512-d).**

- ▶ Identical clustering code, identical K ; only the coordinates change.
- ▶ All four linkages in both spaces: a 2×4 table. Along a row the algorithm changes, down a column the representation changes.
- ▶ Which change moves the ARI more? Does a better representation fix single linkage?

Summary

- K-means and hierarchical clustering: when to apply them, how they work, their pros and cons.
- Hierarchical clustering: the pros and cons of each linkage, and how to use and interpret a dendrogram.
- Clustering learned representations: the choice of space matters as much as the choice of algorithm; DEC learns the space and the clusters jointly.
- Reading: Section 12.4 of [1]. Further reading on deep clustering: [5, 6].
- Acknowledgement:
 - ▶ Figures in this presentation were taken from [1] (Springer, 2023) with permission from the authors: G. James, D. Witten, T. Hastie and R. Tibshirani.
 - ▶ Some of the slides are reproduced based on the slides from T. Hastie and R. Tibshirani.

Hierarchical Clustering: Summary of Linkage Functions

Linkage	Description
Complete	Maximal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>largest</i> of these dissimilarities.
Single	Minimal intercluster dissimilarity. Record the <i>smallest</i> of these pairwise dissimilarities. Single linkage can result in extended, trailing clusters in which single observations are fused one at a time.
Average	Mean intercluster dissimilarity. Record the <i>average</i> of these pairwise dissimilarities.
Centroid	Dissimilarity between the centroid for cluster A and the centroid for cluster B. Can produce inversions, in which clusters fuse below the height of an earlier fusion.
Ward	The increase in the within-cluster sum of squares caused by the merge, $\frac{n_A n_B}{n_A + n_B} \ \mathbf{m}_A - \mathbf{m}_B\ ^2$. Not the same as centroid linkage: it also weights the cluster sizes, and is biased towards balanced, globular clusters.

References

- [1] G. James, D. Witten, T. Hastie, R. Tibshirani and J. Taylor. *An Introduction to Statistical Learning, with Applications in Python*. Springer, 2023. Chapter 12.
- [2] S. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.
- [3] D. Arthur and S. Vassilvskii. k -means++: the advantages of careful seeding. In *Proceedings of SODA*, pages 1027–1035, 2007.
- [4] J. H. Ward. Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association*, 58(301):236–244, 1963.
- [5] J. Xie, R. Girshick and A. Farhadi. Unsupervised deep embedding for clustering analysis. In *Proceedings of ICML*, pages 478–487, 2016.
- [6] S. Zhou, H. Xu, Z. Zheng, J. Chen, Z. Li, J. Bu, J. Wu, X. Wang, W. Zhu and M. Ester. A comprehensive survey on deep clustering: taxonomy, challenges, and future directions. *ACM Computing Surveys*, 57(3), 2025.
- [7] N. Reimers and I. Gurevych. Sentence-BERT: sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, pages 3982–3992, 2019.
- [8] M. Oquab et al. DINOv2: learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.
- [9] L. Hubert and P. Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985.
- [10] K. He, X. Zhang, S. Ren and J. Sun. Deep residual learning for image recognition. In *Proceedings of CVPR*, pages 770–778, 2016.

In-Class Quiz

- The quiz will run from **2:50 pm to 3:50 pm** on **Thursday, 3 September**.
- The quiz must be attempted in person during the Week 6 seminar; attempts outside it will be reported as an academic integrity breach.
- Bring your student ID card (physical card or digital M-Pass) or another form of photo ID for authentication purposes.
- You may bring up to three A4 sheets of notes, double-sided.
- You may use a calculator.
- You are NOT allowed to use any other software (e.g. Python or R), any online services or platforms (e.g. Google, ChatGPT, Claude), or any communication tools (e.g. email, WhatsApp, WeChat) during the quiz.
- Blank paper may be brought for rough work and calculations, but it will not be collected or marked.
- All answers must be entered in Moodle.